BirdGEN: Conditional Bird-Song Spectrogram Generation for Rare Species Classification

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Abstract

BirdGEN investigates whether class-conditional spectrogram generation can improve bird-species classification when labeled data are limited. We use recording-isolated log-mel preprocessing, Residual CNN and CRNN classifiers, and a progressively refined conditional VAE. The downstream study preserves nine matched blocks and the four per-species conditions $50+0$, $50+50$, $50+100$, and $50+200$. The regenerated VAE pools and reused audited Diffusion pools now support current three-seed CRNN-view quality results, and a recorded equal-count benchmark supports the runtime comparison. In the completed low-resource study, validation selected $+200$ generated samples per species for both models. Relative to the 84.75% real-only macro-F1 mean, VAE-v3 reached 87.48% and Diffusion reached 86.21%. Quality, classifier utility, and generator-only spectrogram-sampling time are reported as separate outcomes.

Attestation of Teamwork

Shenghao Jin contributed to data preparation and the conditional VAE. Harvey Li contributed to data preparation, the Residual CNN and CRNN classifiers, the low-resource augmentation experiment, and the VAE–Diffusion generator-only spectrogram-sampling comparison. Vincent Wang developed and evaluated the conditional Diffusion model. All members contributed to the preparation and revision of the final report.

1 Introduction and Problem Formulation

Field recordings are expensive to collect and annotate, so conditional generation may provide additional labeled training examples. Given a species label y and latent variable z, a generator produces a spectrogram $\hat{x} = G(z, y)$. VAEs provide tractable encoding and one-pass decoding [1], while conditional VAEs incorporate the requested class [2]. Although rare-species recognition is the intended application, our experiment simulates *label scarcity* for three project species using generators pretrained on the full training split; it is not an unseen-species experiment.

Our primary question is whether generated samples improve a newly initialized classifier evaluated on fixed, real, held-out recordings. The endpoint is $\Delta F1 = F1_{\text{real+generated}} - F1_{\text{real−only}}$, using macro-F1 to weight the three species equally. The secondary question compares VAE and Diffusion sample quality and equal-count generation time. Quality, downstream utility, and runtime are kept as separate outcomes.

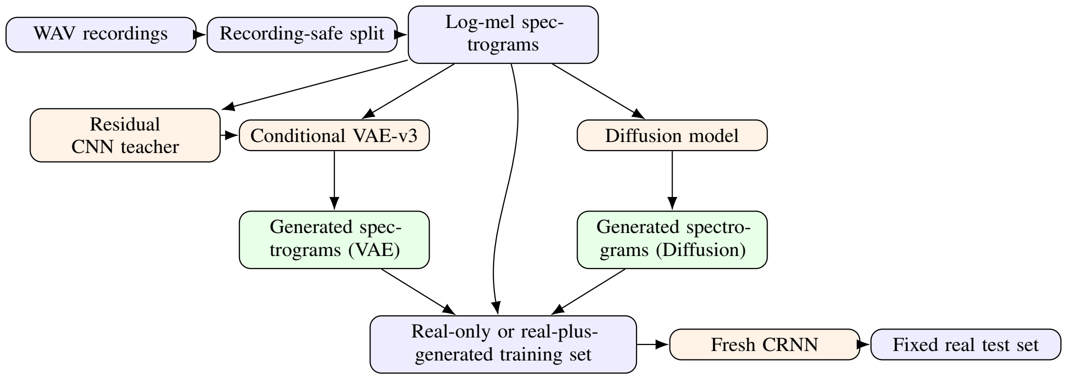

Figure 1: BirdGEN pipeline. Both generated-spectrogram branches feed the same downstream-classifier protocol.

2 Data Preparation

The audited subset contains 3,347 three-second clips from 310 source recording IDs: 1,017 American Robin, 1,074 Northern Cardinal, and 1,256 Song Sparrow. Grouping by recording ID prevents clips from the same recording from crossing train, validation, and test. Historical VAE experiments used 2,339/519/489 clips; the later `content_safe_v2` split removes 24 duplicate training rows and contains 2,315/519/489 clips. Maintained generator comparisons and the refreshed low-resource run use a 510-row generator-safe validation subset, which excludes nine validation rows identified as exact counterparts of historical-training rows. The held-out test set remains unchanged at 489 rows.

Audio is mono at 22.05 kHz and converted to 128×128 log-mel spectrograms using a 1,024-point FFT and hop length 512. The VAE uses training-set global standardization, whereas the maintained classifier maps each clip's relative 80-dB range to $[-1, 1]$; an adapter converts VAE outputs to the classifier scale.

3 System Design and Models

Figure 1 shows the matched evaluation path. Every downstream condition trains a fresh CRNN and uses the same real test set.

3.1 Classifiers

The 1.66-M-parameter Residual CNN uses six residual blocks and global average/max pooling. It is retained as an architecture-comparison reference, and its frozen features and logits guide VAE-v3 training. It does not score or compare VAE and Diffusion in the maintained independent evaluation. The selected 404-k-parameter CRNN combines convolutional spectral features with a bidirectional GRU and temporal pooling [3]. An independent frozen CRNN supplies generation-quality diagnostics, while freshly initialized CRNNs provide the downstream augmentation evidence.

3.2 Conditional VAE Evolution

V1 established a CVAE baseline with a flat 128-value latent, label concatenation, and MSE+KL loss, but its samples blurred narrow calls. V2 used a $16 \times 8 \times 8$ spatial latent, residual FiLM conditioning [4], resize-convolution, detail-aware losses, and species-matched posterior statistics to preserve time–frequency location. V3 increased the latent to $16 \times 16 \times 16$, added free bits and frozen classifier feature/label consistency, and sampled from a species-matched posterior-anchor bank. These changes targeted fine transients, species cues, and the mismatch between the standard-normal prior and the learned posterior. Because several components changed together, the versions are an engineering progression rather than a controlled ablation.

VAE-v3 has 5.37 M parameters. Species embeddings modulate residual blocks with FiLM, and the objective combines event-weighted and multiscale reconstruction, time/frequency gradients, free-bits

KL, and frozen-classifier consistency:

$$\mathcal{L} = \mathcal{L}_{\text{event}} + 0.25\mathcal{L}_{\text{ms}} + 0.60(\mathcal{L}_{\Delta t} + \mathcal{L}_{\Delta f}) + \beta\mathcal{L}_{\text{KL,free}} + s_{\text{cls}}[0.08\mathcal{L}_{\text{feat}} + 0.03\mathcal{L}_{\text{label}}]. \quad (1)$$

Multiscale and feature-space terms are motivated by [5, 7]; free bits follow [6]. For generation, V3 perturbs a class-matched posterior anchor rather than drawing independently from a standard-normal prior, an idea related in motivation to mixture priors such as VampPrior [8].

The corrected reparameterization applies temperature to the stochastic term:

$$z = \mu + 0.35 \ \text{std} \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

Equivalently, the implementation is exactly `z = mu + 0.35*std*epsilon`. The VAE checkpoint was not retrained. Its existing posterior bank was filtered to 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and all three VAE pools were regenerated with this corrected equation. The three audited Diffusion pools were reused without new Diffusion spectrogram generation.

3.3 Diffusion Model

4 Experimental Methodology

Classifier architectures were selected on validation data across seeds 42, 123, and 777, then evaluated on held-out test recordings. VAE evidence is separated into paired deterministic reconstruction, unpaired generated-sample quality, and downstream classifier utility. To address the presentation feedback, each frozen generated sample was compared with its nearest same-species test spectrogram in the standardized-logmel domain, with a leave-one-out real-test reference. The test set was used only for this final evaluation; low MSE can also indicate copying or limited diversity.

For the primary experiment, each fresh CRNN sees 50 real examples per species. Three real-subset seeds crossed with three classifier seeds give nine matched blocks. Every block evaluates $50 + 0$, $50 + 50$, $50 + 100$, and $50 + 200$ real-plus-generated samples per species, separately for VAE and Diffusion. Ratios are selected using the 510-row generator-safe validation subset; only the shared $50 + 0$ baseline and each generator's selected ratio reach the fixed 489-row real test set. Architecture, augmentation, and optimizer-step budget are matched. No existing pretrained classifier is augmented, and the Residual CNN is not a generated-sample evaluator. The recorded runtime benchmark timed five fresh in-memory 600-spectrogram repeats per model on the same GPU, precision, and batch size after warm-up. CUDA was synchronized around generator sampling through the final normalized tensor; model loading, existing arrays, file output, plotting, and metric computation were excluded. Models were measured sequentially: the completed Diffusion benchmark was retained and VAE-v3 was remeasured with the filtered bank, with no Diffusion sampling rerun.

5 Experimental Results

Classifier and VAE diagnostics. The selected CRNN obtained 90.16% macro-F1 on 489 held-out clips. Across VAE versions, deterministic test reconstruction MSE decreased from 0.1533 (V1) to 0.0654 (V2) and 0.0281 (V3); V3 test MAE was 0.1261 and spectral convergence was 0.1794. These paired reconstruction values do not measure the distance between independently generated and real examples.

The 48 historical samples achieved 100% target agreement under the Residual teacher (mean confidence 0.915), but that classifier also contributes to Eq. (1); the result is not an independent realism score. For the separately verified seed-42 notebook pool (16 samples/species), the same-species generated-to-test nearest-neighbor MSE was 0.5120 on average (SD 0.1998; median 0.4501; IQR 0.3878–0.6231), versus 0.3350 for the leave-one-out real-to-real reference (SD 0.1419; median 0.3185; IQR 0.2368–0.4154). Generated/real means were 0.4308/0.3237 for American Robin, 0.5579/0.3650 for Northern Cardinal, and 0.5474/0.3175 for Song Sparrow. These are unpaired pixel-space summaries (48 versus 489 queries), not a perceptual-realism or waveform-quality score; nearest-neighbor MSE can reward copying. This hash-recorded diagnostic used the original notebook sampler at temperature 0.35 with a train-only posterior bank, not Harvey's corrected canonical pool. The Residual CNN remains a frozen VAE training teacher and supplies no independent VAE–Diffusion evaluation. Table 1 reports the current audited pools using the selected CRNN and its

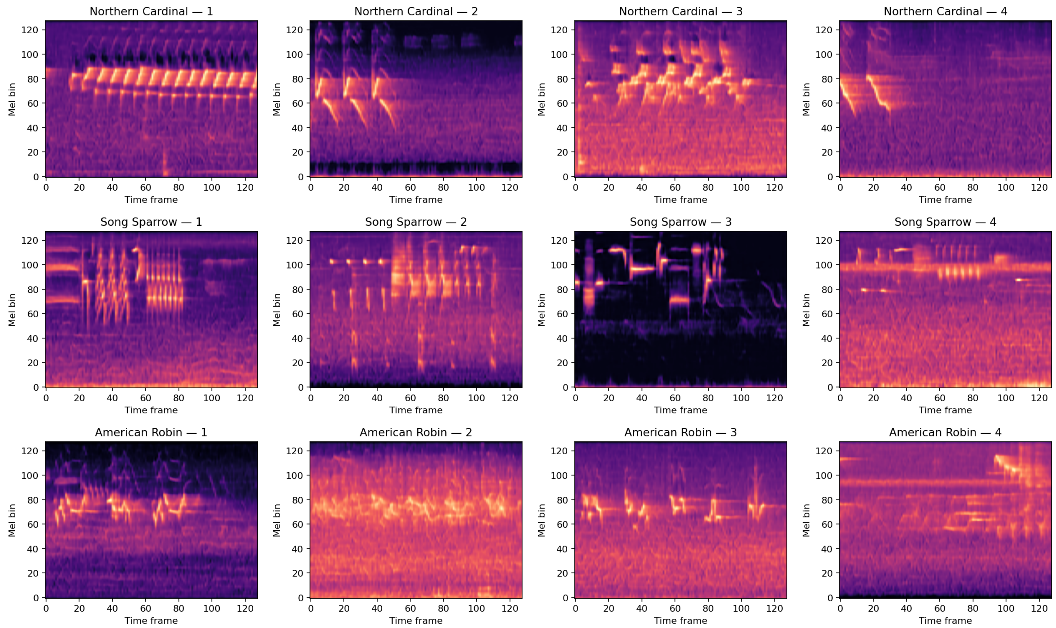

Figure 2: Existing VAE-v3 notebook posterior-anchor samples at temperature 0.35: four examples per species (48 samples were saved in total). Panels are independently color-normalized; spectrogram appearance alone does not establish waveform realism.

Table 1: Frozen selected-CRNN and feature-space generated-sample diagnostics.

Metric	VAE-v3	Diffusion
Target-label accuracy / CRNN macro-F1	96.67% / 96.66%	92.44% / 92.42%
Classifier-feature Fréchet distance	22.8950	41.2550
Feature precision / feature recall	0.8528 / 0.8972	0.5654 / 0.8207
Feature density / coverage / diversity ratio	0.6423 / 0.7419 / 0.8719	0.3114 / 0.4845 / 1.0083
Pixel Wasserstein / near-train fraction / exact copies	0.0787 / 36.67% / 0	0.1916 / 2.50% / 0

shared feature space. Values are three-generation-seed means; the feature rows additionally average the three species.

Low-resource augmentation. Test reporting is limited to the shared $50 + 0$ baseline and the independently selected $+200$ condition. Mean held-out macro-F1 was 84.75% for real-only, 87.48% for VAE-v3, and 86.21% for Diffusion. The paired changes were $+2.72$ percentage points for VAE-v3 (7/9 positive blocks; descriptive block-bootstrap interval $+1.00$ to $+4.65$) and $+1.46$ points for Diffusion (6/9; $+0.15$ to $+2.89$).

Equal-count generator timing. The recorded benchmark generated 600 fresh in-memory spectrograms (200 per species) per synchronized repeat after warm-up. The package retains every repeat's seconds, throughput, peak accelerator memory, hardware, precision, batch size, sampler settings, seed, measurement sequence, and timing boundary. Pre-existing arrays are not timing evidence. The median [Q1, Q3] repeat time was 0.3046 [0.2983, 0.3076] seconds for VAE-v3 and 411.1814 [410.9396, 411.2319] seconds for Diffusion. These measurements use FP32, batch size 8, five warm-up batches, five CUDA-synchronized repeats, and an RTX 4070 SUPER. They exclude loading, conversion, transfer, file I/O, and audio decoding. The completed Diffusion benchmark was retained; VAE-v3 was remeasured afterward using the filtered 759-anchor bank, and no Diffusion sampling was rerun.

Table 2: Completed low-resource CRNN study across nine matched blocks.

Per-species condition	Val. macro-F1 (VAE / Diff.)	Test status
$50 + 0$	Shared real-only reference	Evaluated baseline
$50 + 50$	84.53% / 85.45%	Not selected
$50 + 100$	85.88% / 86.06%	Not selected
$50 + 200$	86.47% / 86.84%	Selected for both

Table 3: Generator-only timing comparison for matched fresh outputs.

Model	Outputs/repeat	Repeat seconds	Samples/s	Peak accelerator memory
VAE-v3 (corrected sampler)	600	0.3058 ± 0.0096	1963.276	0.185 GiB
Diffusion (100-step DDIM)	600	412.3090 ± 2.8120	1.455	0.333 GiB
Difference / Diffusion-to-VAE ratio	600	412.0032 / 1348.09×	–	–

6 Discussion

In the simulated low-resource setting, both validation-selected $+200$ arms improved mean held-out macro-F1 versus their matched real-only controls. The nine blocks reuse overlapping real subsets, so the intervals are descriptive and the largest tested ratio is not an estimated optimum. This result applies to newly initialized low-resource CRNNs, not the historical full-data CRNN. Reconstruction MSE rewards pixel proximity, the Residual CNN is coupled to V3 training and therefore restricted to its teacher role, and a closed-set independent CRNN cannot reject unknown or noisy inputs. Spectrogram and classifier metrics also do not replace listening or waveform-level evaluation [9]. VAE-v3 has higher CRNN target compatibility, lower macro-species Fréchet distance, and higher feature precision/coverage in these pools, but it also has a much larger near-training feature fraction. Diffusion has a diversity ratio nearer one and lower copy-risk flags. Separately, Diffusion took $1348.09\times$ the VAE-v3 generator time under the recorded boundary.

7 Conclusion

BirdGEN now has corrected three-seed generator evidence, a completed matched low-resource CRNN study, and a matched timing comparison. Validation selected +200/species for both models; the mean paired macro-F1 changes were $+2.72$ points for VAE-v3 and $+1.46$ points for Diffusion. Quality diagnostics remain metric-specific rather than a composite winner, and the efficiency result is limited to generator-only spectrogram sampling on the recorded hardware.

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A Supplementary Reproducibility Notes

VAE-v3 used AdamW with learning rate 2×10^{-4}, batch size 24, a 20-epoch KL warm-up, and checkpoint selection by the complete validation objective. The portable pool generator now applies the notebook's sampling temperature 0.35 explicitly. The VAE checkpoint was not retrained. The existing posterior bank was filtered to 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and the three canonical VAE pools were regenerated with the corrected sampler. The reported generated-to-test MSE instead uses a separate, hash-recorded seed-42 pool reconstructed with the original notebook, 16 samples per species, and the train-only posterior bank. This MSE package remains separate from the three-seed augmentation pools and does not replace their quality or downstream CRNN evidence.

For the equal-count timing study, each model generated 600 spectrograms (200 per species) in FP32 with batch size 8 on the same hardware. Five warm-up batches preceded five synchronized repeats. Models were measured sequentially, not interleaved: the completed Diffusion benchmark was retained and VAE-v3 was remeasured after the bank filter. The sole timing boundary is generator sampling through the final normalized spectrogram tensor; model loading, existing arrays, file output, plotting, and metric computation are excluded. The package preserves repeat seconds, throughput, peak accelerator memory, hardware, precision, batch size, seeds, sampler settings, and the timing-boundary description.